

Cruise Control System – PID Controller Simulation in MATLAB

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1. Introduction

This project simulates a basic cruise control system using MATLAB and Simulink. The objective is to design, implement, and evaluate different controllers—P and PID—to improve system stability and performance.

Plant model:

$$G(s) = 1 / (1000s + 50)$$

Tools Used:

- MATLAB
- Simulink
- Control System Toolbox

2. Controller Design

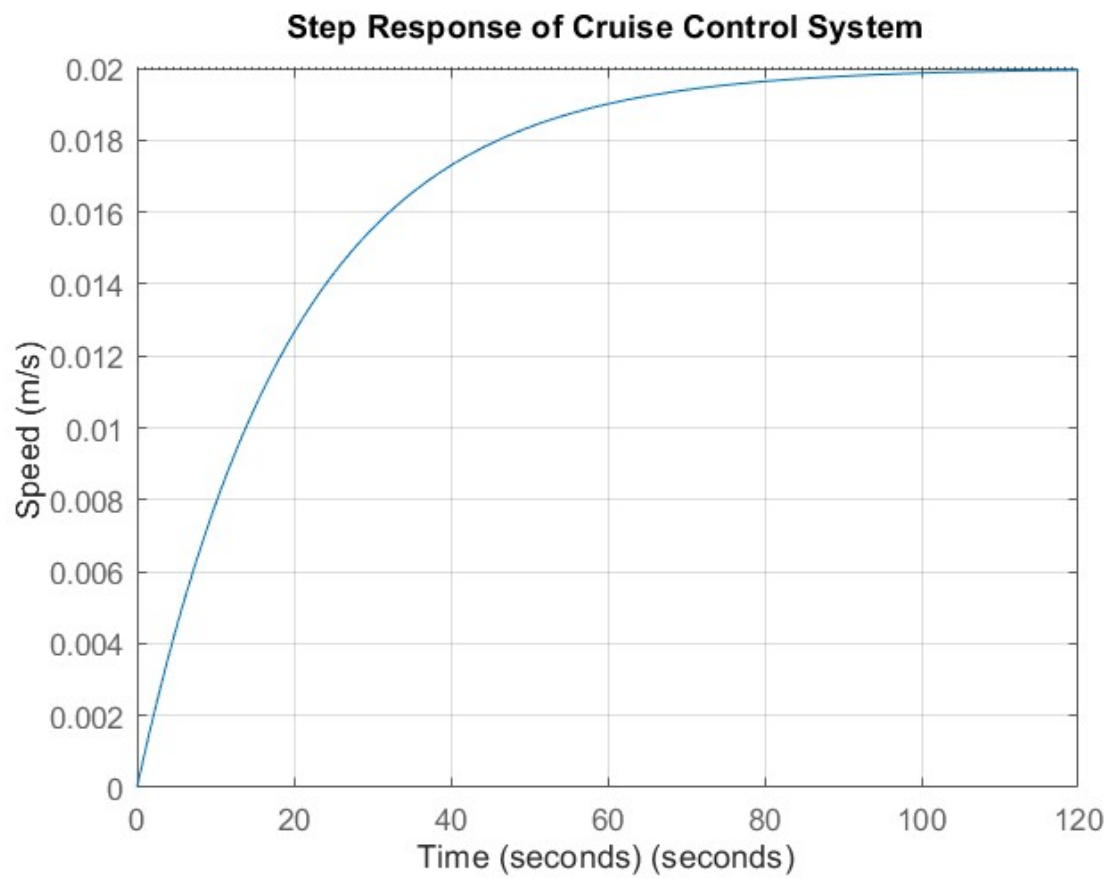
We began by applying a simple proportional controller, followed by a full PID controller. The tuning of the PID parameters was performed manually to achieve desired rise time, overshoot, and steady-state error.

3. System Performance Metrics

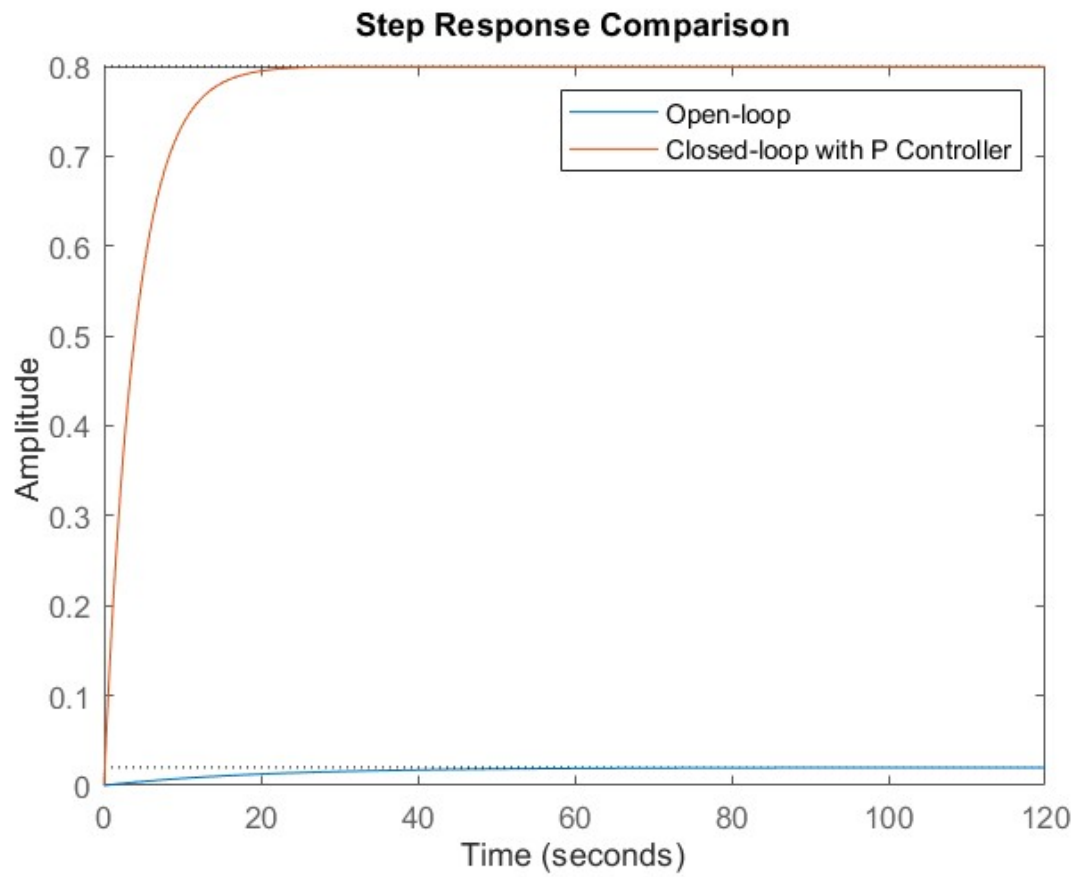
Controller Type	Rise Time	Settling Time	Overshoot	Steady-State Error
P Controller	~13 s	~50 s	~7.9%	Present
PID Controller	~169 s	~480 s	0%	Nearly zero

4. Simulation Plots

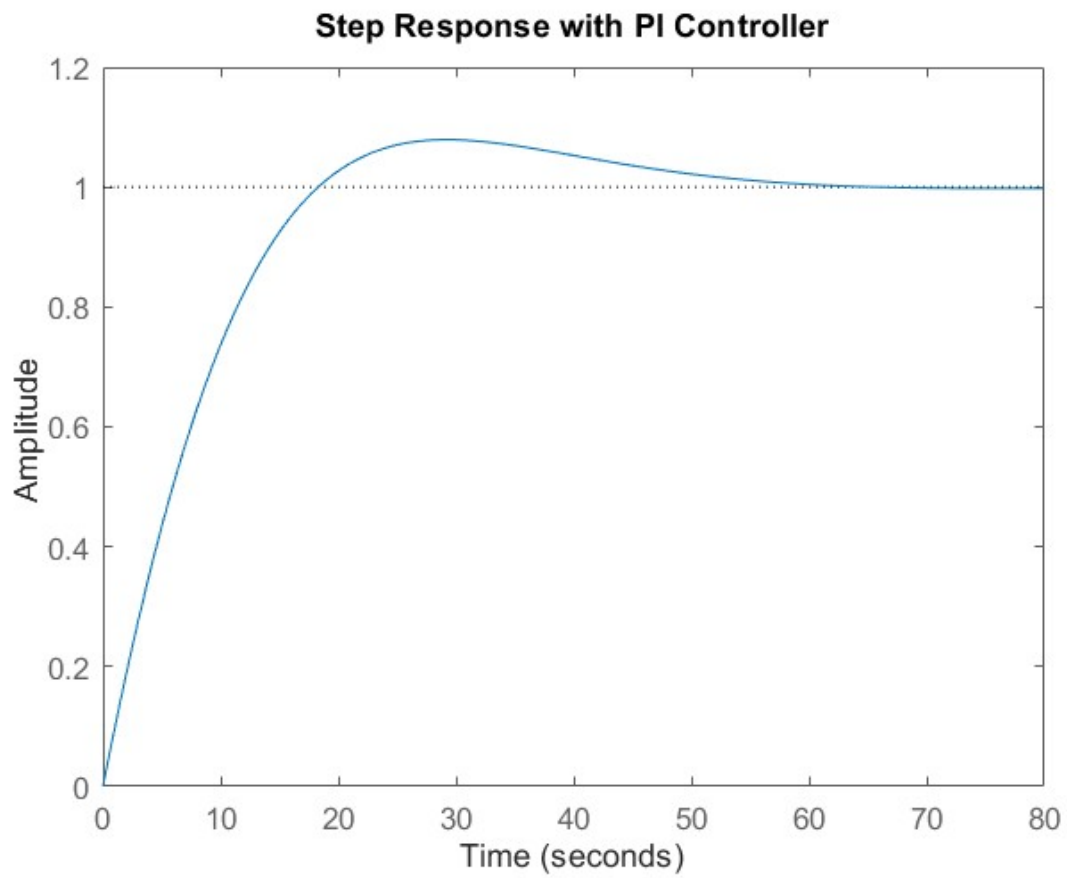
4.1 Step Response – Open Loop System



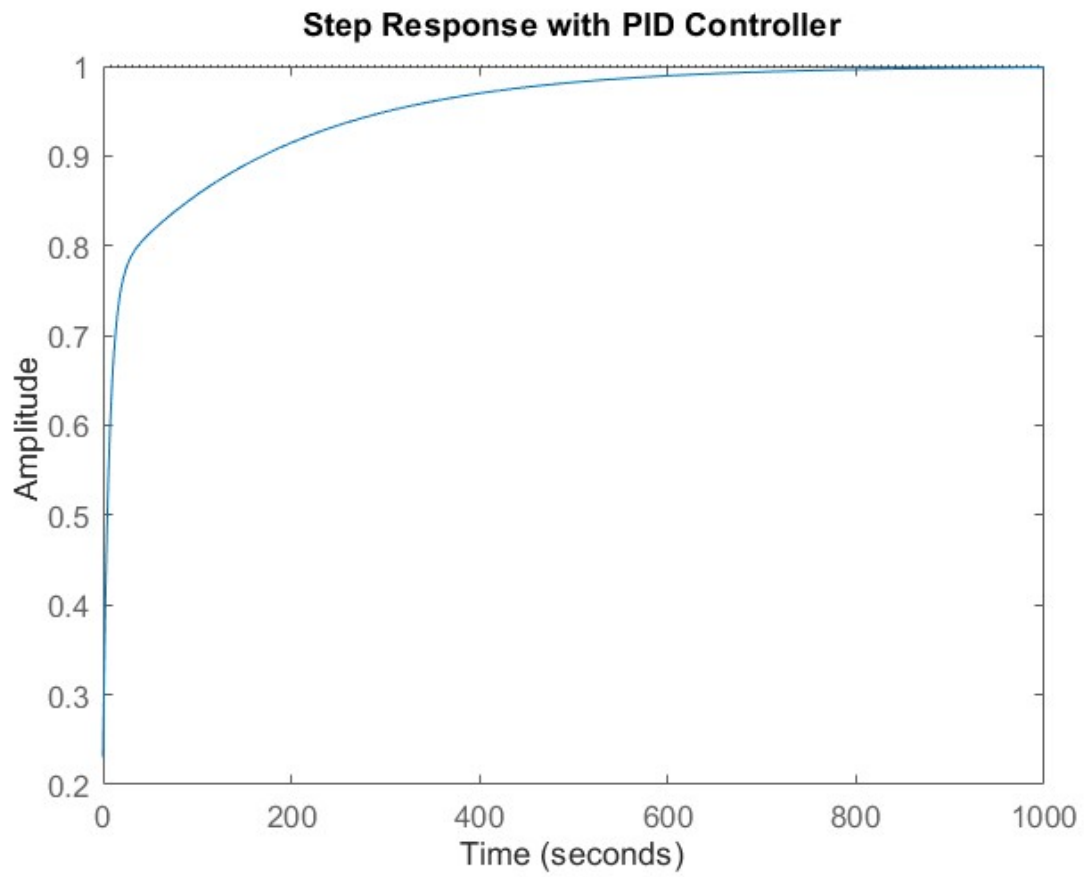
4.2 Step Response – With P Controller



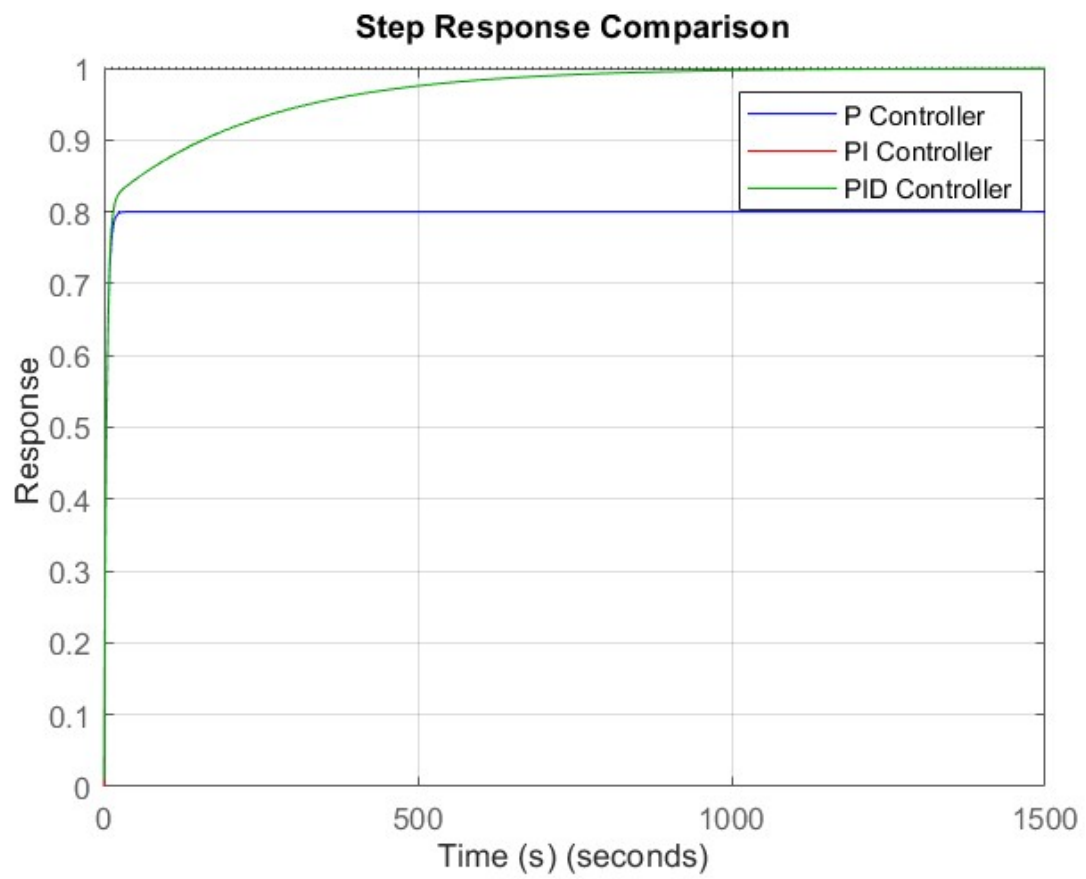
4.3 Step Response – With PI Controller



4.4 Step Response – With PID Controller



4.5 Controller Comparison



5. Discussion

The open-loop system responds slowly and never reaches the desired speed. With proportional control, performance improves significantly but causes some overshoot. A more tuned controller achieves accurate tracking with no overshoot but at the cost of a slower response.

6. Conclusion

The cruise control system's stability and accuracy significantly improve with the implementation of a tuned PID controller. This simulation highlights the importance of controller selection and tuning in practical control systems.

Future Work:

- Apply to nonlinear or real-time models
- Extend to adaptive or robust control strategies